

Computer Assignment

Pointers

Program 1: Pointers and Array

```
#include <stdio.h>

int main(){
    int *pt;

    printf("Enter No of elements on an array: ");
    int n;
    scanf("%d",&n);

    int i,arr[n];
    for(i = 0; i<n ; i++){
        printf("Element %d: ",i+1);
        scanf("%d", &arr[i]);
    }
    //displaying array using pointer
    pt = arr;
    printf("\nStored array: ");
    for(i = 0; i<n ; i++){
        printf("%d  ",*(pt+i));
    }

    return 0;
}
```

Output:

```
Enter No of elements on an array: 4
Element 1: 1
Element 2: 3
Element 3: 4
```

Element 4: 2

Stored array: 1 3 4 2

Program 2 : Pointer Passed in UDF

```
#include <stdio.h>

void passpointer(int *);

int main(){
    printf("Enter a Number: ");
    int num;
    scanf("%d",&num);
    int *pt = &num;
    passpointer(pt);
    return 0;
}

void passpointer(int *ptr){
    printf("\n\nMemory Passed: %d\nValue: %d",ptr,*ptr);
}
```

Output:

Enter a Number: 234

Memory Passed: 2043108860
Value: 234

Enter a Number: 546

Memory Passed: 27997676
Value: 546

Program 3: Pointer and Structure

```
#include <stdio.h>

struct company {
    char name[30];
    char location[30];
    int No_Of_Employee;
    float annual_budget;
};

void readStruct(struct company *);
void printStruct(struct company *);

int main(){
    struct company C;
    readStruct(&C);
    printStruct(&C);
    return 0;
}

void readStruct(struct company *C){
    printf("Enter Following Details of Company: \n");
    printf("Name: ");
    scanf(" %[^\\n]s",C->name);
    printf("Location: ");
    scanf(" %[^\\n]s",C->location);
    printf("Number of Employees: ");
    scanf("%d",&C->No_Of_Employee);
    printf("Annual Budget: ");
    scanf("%f",&C->annual_budget);
}

void printStruct(struct company *C){
    printf("\n----Stored Data----\nName: %s\nLocation: %s\n\n");
    printf("No of Employees: %d\nAnnual Budget: %.2f",C->name,C->locati
```

```
on,C->No_Of_Employee,C->annual_budget);  
}
```

Output:

Enter Following Details of Company:

Name: Chaudhari Group

Location: Satungal, Kathmandu

Number of Employees: 3254

Annual Budget: 87453766

----Stored Data----

Name: Chaudhari Group

Location: Satungal, Kathmandu

No of Employees: 3254

Annual Budget: 87453768.00

Structures

Program 1: Normal single Structure

```
#include <stdio.h>  
  
struct company {  
    char name[30];  
    char location[30];  
    int No_Of_Employee;  
    float annual_budget;  
};  
  
int main(){  
    struct company C;  
    printf("Enter Following Details of Company: \n");  
    printf("Name: ");  
    scanf(" %[^\\n]s",C.name);  
    printf("Location: ");
```

```

scanf(" %[^\\n]s",C.location);
printf("Number of Employees: ");
scanf("%d",&C.No_Of_Employee);
printf("Annual Budget: ");
scanf("%f",&C.annual_budget);

//Displaying Structure
printf("\\n----Stored Data----\\nName: %s\\nLocation: %s\\n
No of Employees: %d\\nAnnual Budget: %.2f",C.name,C.location,
C.No_Of_Employee,C.annual_budget);

return 0;
}

```

Output:

```

Enter Following Details of Company:
Name: Nischhal ko Khatra Company
Location: mato ma xa
Number of Employees: 1
Annual Budget: 0.222

----Stored Data----
Name: Nischhal ko Khatra Company
Location: mato ma xa
No of Employees: 1

```

Program 2: Array of Structure

```

#include <stdio.h>

struct company {
    char name[30];
    char location[30];
    int No_Of_Employee;
    float annual_budget;
};

```

```

int main(){
    printf("Enter Number of Company: ");
    int n;
    scanf("%d",&n);

    struct company C[n];
    int i;
    printf("Enter Following Details:: \n");
    for(i = 0 ; i<n ; i++){
        printf("----For Company %d---- \n",i+1);
        printf("Name: ");
        scanf(" %[^\\n]s",C[i].name);
        printf("Location: ");
        scanf(" %[^\\n]s",C[i].location);
        printf("Number of Employees: ");
        scanf("%d",&C[i].No_Of_Employee);
        printf("Annual Budget: ");
        scanf("%f",&C[i].annual_budget);
        printf("\\n");
    }

    //Displaying Structure
    printf("\\n----Stored Data----\\n");
    printf("\\nName\\t\\t\\tLocation\\t\\tNo of Employees\\t\\tAnnual Budget\\n");
    for(i=0 ; i<n ; i++)
        printf("%-16s\\t%-16s\\t%-24d%- .2f\\n",C[i].name,C[i].location,C[i].No_Of_Employee,C[i].annual_budget);
    return 0;
}

```

Output:

Enter Number of Company: 5

Enter Following Details::

----For Company 1----

Name: Nischhal Pvt Limited
Location: Chandragiri
Number of Employees: 324
Annual Budget: 232523

----For Company 2----

Name: Clamphook Acadamy
Location: Thapata;i
Number of Employees: 23
Annual Budget: 234

----For Company 3----

Name: PEA
Location: Thapathali
Number of Employees: 23
Annual Budget: 342325

----For Company 4----

Name: Tesr
Location: Test
Number of Employees: 334
Annual Budget: 3434

----For Company 5----

Name: Test 2
Location: ldsfds
Number of Employees: 34
Annual Budget: 25

----Stored Data----

Name	Location	No of Emplo
yees	Annual Budget	
Nischhal Pvt Limited	Chandragiri	324
232523.00		
Clamphook Acadamy	Thapata;i	23
234.00		

PEA	Thapathali	23
342325.00		
Tesr	Test	334
3434.00		
Test 2	ldsfds	34
25.00		

File Handling

Program 1: fputc() and fgetc()

```
#include <stdio.h>

int main(){
    printf("Enter a File Name: ");
    char filename[20];
    gets(filename);
    FILE *fp = fopen(filename,"w");
    printf("Enter a File Content: ");
    char ch;
    while((ch = getchar())!='\n'){
        fputc(ch,fp);
    }
    printf("\nFile sucessfully Written.....\n");
    fclose(fp);
    fp = fopen(filename,"r");
    printf("\nReading file.....\n");
    printf("File content:\n_____ \n");
    //ch = fgetc(fp);
    //while(ch!=EOF){
    while((ch = fgetc(fp))!=EOF){
        printf("%c",ch);
        //ch = fgetc(fp);
    }
    printf("\n_____ \n");
    fclose(fp);
}
```



```
    return 0;
}
```

Output:

```
Enter a File Name: Hello.txt
Enter a File Content: Hello How you doing broo suppp

File sucessfully Written.....

Reading file.....
File content:
_____
Hello How you doing broo suppp
_____
```

Program 2: fputs() and fgets()

```
#include <stdio.h>

int main(){
    printf("Enter a File Name: ");
    char filename[20];
    gets(filename);
    FILE *fp = fopen(filename,"w");
    printf("Enter a File Content: ");
    char str[300];
    scanf(" %[^\\n]s",str);
    fputs(str,fp);

    printf("\\nFile sucessfully Written.....\\n");
    fclose(fp);
    fp = fopen(filename,"r");
    printf("\\nReading file.....\\n");
    printf("File content:\\n_____\\n");
    while(fgets(str,300,fp)){
        printf("%s",str);
    }
}
```

```

    }
    printf("\n_____ \n");
    fclose(fp);

    return 0;
}

```

Output:

```

Enter a File Name: kxa.txt
Enter a File Content: k xa vai sanchai chas ni??

File sucessfully Written.....

Reading file.....
File content:
_____
k xa vai sanchai chas ni??
_____

```

Program 3: fprintf() and fscanf()

```

#include <stdio.h>

int main(){
    printf("Enter a File Name: ");
    char filename[20];
    gets(filename);
    FILE *fp = fopen(filename,"w");
    char name[20];
    char adress[20];
    char gender;
    char DOB[12];
    printf("Enter Number of people: ");
    int n;
    scanf("%d",&n);
    int i;

```

```

printf("Enter following Details:\n");
for(i=0 ; i<n ; i++){
    printf("\nPerson %d:\n",i+1);
    printf("Name: ");
    scanf(" %[^\\n]s",name);
    printf("Adress: ");
    scanf(" %[^\\n]s",adress);
    printf("Gender: ");
    scanf(" %c",&gender);
    printf("DOB: ");
    scanf(" %s",DOB);
    //writing file in a formatted form using fprintf
    fprintf(fp,"%s,%s,%c,%s\\n",name,adress,gender,DOB);
}
printf("\nFile sucessfully Written.....\\n");
fclose(fp);
fp = fopen(filename,"r");
printf("\nReading file.....\\n");
printf("File content:\\n_____\\n");
//reading file using fscanf()
while (fscanf(fp, " %[^,],%[^,],%c,%s", name, adress, &
gender, DOB) == 4) {
    printf("%s, %s, %c, %s\\n", name, adress, gender, DO
B);
}
printf("_____\\n");
return 0;
}

```

Output:

```

Enter a File Name: test.txt
Enter Number of people: 3
Enter following Details:

Person 1:
Name: Nischhal Shrestha
Adress: Chandragiri

```

Gender: M
DOB: 2064-05-01

Person 2:
Name: Nikit Bhatta
Adress: Lalitpur
Gender: M
DOB: 2066-04-09

Person 3:
Name: Mahodaya Bhandari
Adress: Deauba Chowk
Gender: M
DOB: 2009-88-08

File sucessfully Written.....

Reading file.....
File content:

Nischhal Shrestha, Chandragiri, M, 2064-05-01
Nikit Bhatta, Lalitpur, M, 2066-04-09
Mahodaya Bhandari, Deauba Chowk, M, 2009-88-08

Program 4: fread() and fwrite()

```
#include <stdio.h>

struct person{
    char name[20];
    char adress[20];
    char gender;
    char DOB[12];
};

int main(){
```

```

struct person P;
printf("Enter a File Name: ");
char filename[20];
gets(filename);
FILE *fp = fopen(filename,"w");
printf("Enter Number of people: ");
int n;
scanf("%d",&n);
int i;
printf("Enter following Details:\n");
for(i=0 ; i<n ; i++){
    printf("\nPerson %d:\n",i+1);
    printf("Name: ");
    scanf(" %[^\\n]s",P.name);
    printf("Adress: ");
    scanf(" %[^\\n]s",P.adress);
    printf("Gender: ");
    scanf(" %c",&P.gender);
    printf("DOB: ");
    scanf(" %s",P.DOB);
    //writing file in a formatted form using fprintf
    fwrite(&P,sizeof(struct person),1,fp);
}
printf("\nFile sucessfully Written.....\n");
fclose(fp);
fp = fopen(filename,"r");
printf("\nReading file.....\n");
printf("File content:\n_____\\n");
//reading file using fscanf()
while (!feof(fp)) {
    fread(&P,sizeof(struct person),1,fp);
    printf("%s, %s, %c, %s\\n", P.name, P.adress, P.gend
er, P.DOB);
}
printf("_____\\n");

return 0;
}

```

Output:

Enter a File Name: data.txt

Enter Number of people: 2

Enter following Details:

Person 1:

Name: ram shrestha

Adress: somewhere

Gender: M

D0B: 232-45-543

Person 2:

Name: Mayalu Karki

Adress: jere tole

Gender: F

D0B: 234-545-443

File sucessfully Written.....

Reading file.....

File content:

ram shrestha, somewhere, M, 232-45-543

Mayalu Karki, jere tole, F, 234-545-443
